Applications of Pythagoras' theorem

ANSWER KEY

Core Concepts: Applications of Pythagoras' theorem

1 In $\triangle ABC$ studying applications of pythagoras' theorem, $m\angle A = 38^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 97° ✓

b. 107°

c. 89°

d. 142°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (38^\circ + 45^\circ) = 97^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm ✓

c. 21 cm

d. 28 cm

Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 16

c. 14 ✓

d. 19

Substituting $x = 1$ into $y = 5x + 9$ gives $y = 5(1) + 9 = 14$.

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying applications of pythagoras' theorem, $m\angle A = 38^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 90° ✓

b. 100°

c. 82°

d. 142°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (38^\circ + 52^\circ) = 90^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm ✓

c. 28 cm

d. 63 cm

Multiplying the original length by the scale factor gives $7 \times 3 = 21$ cm.

6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$?

a. 14

b. 21

c. 19 ✓

d. 24

Substituting $x = 2$ into $y = 5x + 9$ gives $y = 5(2) + 9 = 19$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying applications of pythagoras' theorem, $m\angle A = 38^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 83° ✓

b. 93°

c. 75°

d. 142°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (38^\circ + 59^\circ) = 83^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm ✓

c. 35 cm

d. 112 cm

Multiplying the original length by the scale factor gives $7 \times 4 = 28$ cm.

9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$?

a. 19

b. 26

c. 24 ✓

d. 29

Substituting $x = 3$ into $y = 5x + 9$ gives $y = 5(3) + 9 = 24$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying applications of pythagoras' theorem, $m\angle A = 38^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 76° ✓

b. 86°

c. 68°

d. 142°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (38^\circ + 66^\circ) = 76^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm ✓

c. 42 cm

d. 175 cm

Multiplying the original length by the scale factor gives $7 \times 5 = 35$ cm.

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 4$?

a. 24

b. 31

c. 29 ✓

d. 34

Substituting $x = 4$ into $y = 5x + 9$ gives $y = 5(4) + 9 = 29$.

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